Evaluating Aerosol Effects on MethaneSAT's CO_2 Retrieval

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Introduction

- **Carbon dioxide** (CO_2) emissions are a major driver of **global warming**, and improved monitoring is essential for informing climate policy.
- Can we use **MethaneSAT**, a satellite designed to estimate methane (CH_4) emissions, to **estimate CO_2 emissions**?
- MethaneSAT has two **spectrometers** on board, which retrieve at the following three wavelengths: 1286 nm, which is used to measure oxygen (O_2) concentration; 1611 nm, which is used to measure CO_2; and 1640 nm, which is used to measure CH_4.
- Retrieved signals can be influenced by phenomena such as **Rayleigh scattering**, **single-scattering albedo from aerosols**, **sudden variations in albedo**, **clouds**, etc.
- To correct the CO_2 product, an O_2 proxy was used. We assessed its effectiveness.
- Surface pressure is calculated from the O_2 column estimate.



MethaneSAT (from methanesat.org)

Methods

Data from **MethaneSAT** was compared with data from the Visible Infrared Imaging Radiometer Suite (**VIIRS**). We ran robust regressions in R between the following:

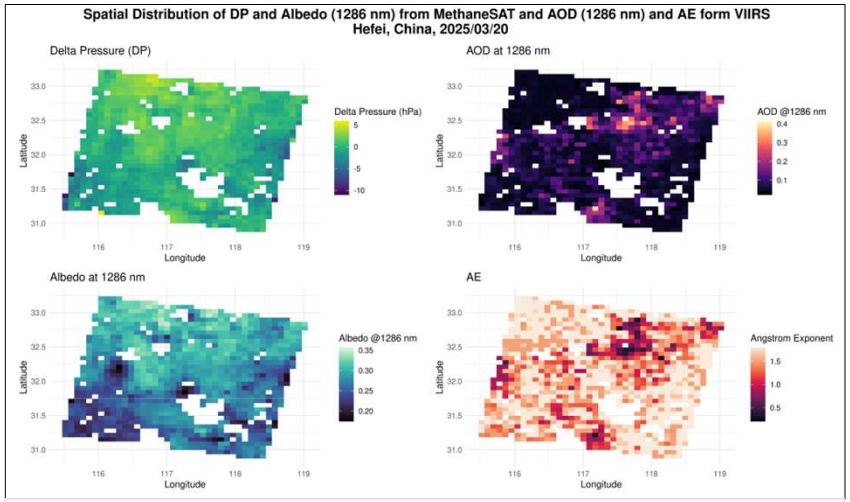
- **dP** from MethaneSAT (i.e. retrieved surface pressure minus prior surface pressure) and Aerosol Optical Density (**AOD**) at 1286nm (calculated using the Ångström-power law) from VIIRS.
- **dP** from MethaneSAT and **albedo** at 1286nm from MethaneSAT, (**AE**) from VIIRS and **AOD** at 1286nm (calculated using the Ångström-power law) from VIIRS.

$$O_2\text{-proxy: } XCO_2 = \frac{N_{CO_2}}{N_{O_2}} \cdot XO_{2,0}$$

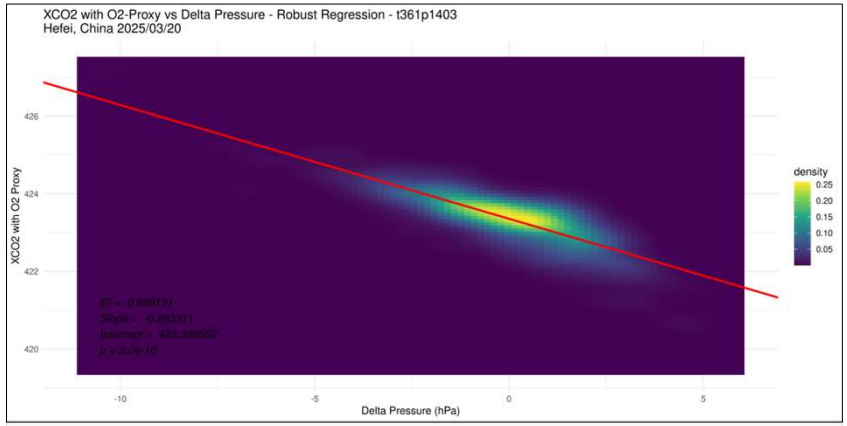
Acknowledgements
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Results

The regression **between dP and AOD (1286 nm) + albedo (1286 nm) + AE**, showed a better average pseudo-R^2 and pseudo-p-value when looking at multiple scenes. This is probably due to albedo being retrieved by MethaneSAT instead of VIIRS, which would reduce the effect of wind transport and different timestamps in the retrieval (MethaneSAT's overpass time is around 13:30 local time, whereas VIIRS's is at around 15:00 local time).



Plots showing the distribution of the regressed variables over the same scene

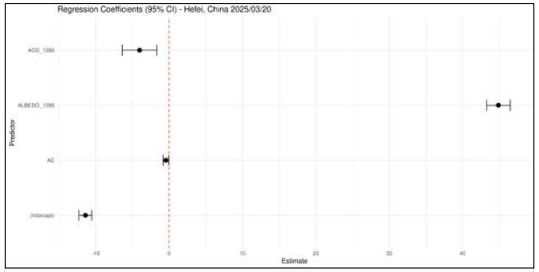


Density plot showing correlation between CO_2 and dP

Miller, Christopher Chan, et al. *Methane Retrieval from MethaneAIR Using the CO_2 Proxy Approach: A Demonstration for the Upcoming MethaneSAT Mission*. *Atmospheric Measurement Techniques*, vol. 17, no. 17, 2024, pp. 5429–5450. Copernicus Publications, https://doi.org/10.5194/amt-17-5429-2024.
Lee, Jaehwa, et al. *VIIRS Version 2 Deep Blue Aerosol Products*. *Journal of Geophysical Research: Atmospheres*, vol. 129, no. 6, 2024, e2023JD040082. Wiley, https://doi.org/10.1029/2023JD040082.
https://www.methanesat.org

Conclusion

The correlation between dP and the other variables suggests that **retrieved pressure is strongly influenced by albedo, aerosols, and AE, undermining the accuracy of the O_2 proxy**. While this doesn't rule out a CO_2 product from MethaneSAT, it highlights the need for a **new correction method**.



Effects of each parameter on dP visualized in scenes over Hefei, China, 2025/03/20

Next Steps

- Devise a new proxy that more rigorously corrects the CO_2 output penalizing the retrieved value when the dP is high, thus **minimizing the influence of albedo and aerosols on the result**.
- Devise a **secondary correction** that takes out albedo and aerosol effects.
- **Compare the CO_2 data from MSAT with data from other remote sensing systems** (i.e. OCO-3, OCO-2, GOSAT, CAMS) to try to see if there is some other parameter for which we can correct the MethaneSAT CO_2 output more directly.
- Run regressions **by aerosol type** (i.e. background, mixed, smoke, etc.), instead of by scene, to try to find patterns in correlation with dP.

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